RESEARCH REPORT · QUANTUM COMPUTING RELIABILITY

WARNINGS A QUBIT MONITOR CAN STAND BEHIND

An assessment of Quantum Noise Profiler (qnp), an open-source tool that detects qubit instability, controls how often it raises false alarms over a whole monitoring session, and refuses to report measurements its data cannot support.

SCOPE OF EVIDENCE All results in this report come from two preregistered simulation studies (Qiskit Aer, Cirq, numpy). It contains no measurements from real quantum hardware.

DOCUMENT CONTROL

Prepared by	Christopher Woodyard
Organization	Vers3Dynamics
Software	qnp 0.3.0.dev0, Python, Qiskit and Cirq adapters
Licence	MIT, open source
Evidence	Two preregistered simulation studies
Hardware data	**None yet; 14-job pilot preregistered, not run**
Date	13 October 2026
Repository	topherchris420/quantum-noise-profiler on GitHub

CONTENTS

00	Summary	2
01	Evidence status	3
02	Background	3
03	Capabilities	4
04	Results, v0.1 to v0.3	5
05	Against standard methods	6
06	Measurements it refuses to report	7
07	Do earlier warnings help?	8
08–09	What is not known · Engagement	9
10	Quick start · References	10

00 · SUMMARY

SUMMARY FOR DECISION-MAKERS

WHAT IT IS

A monitor for qubit error rates and coherence times between calibrations. Every alert states the evidence and the kind of statistical guarantee behind it; estimates the data cannot support are withheld with a stated reason.

WHAT HAS BEEN SHOWN

In two preregistered simulation studies, 53 of 62 and then 40 of 40 criteria passed. Session-level false alerts on stable qubits fell from 0.43–0.88 under v0.1 to at most 0.010 with anytime-valid thresholds.

WHAT HAS NOT

No real-device data exist. In the operational simulation, a calibrated CUSUM chart did as well as the profiler's engine. Interventions available to cloud users, such as remapping away from a degraded qubit, are untested.

WHAT IS BEING ASKED

Independent replication, and partners for a proposed 30-day, read-only observation study on existing calibration records. That phase submits no jobs and changes no device settings (page 9).

TABLE 1 · HEADLINE FINDINGS — SIMULATION · V0.1 → V0.3

01	Stable 8 h sessions that raised any false alert (qubit at 4.5%, threshold 5%)	0.70 → 0.010
02	T1 reported confidently wrong under slow readout drift during the sweep (v0.2 → v0.3)	0.255 → 0.005
03	Median warning before a drifting qubit crosses 5%, at a matched false-alert rate (vs Shewhart / CUSUM)	253 vs 223 / 213 min
04	Jobs run on a degraded qubit: daily calibration alone vs adding recalibration on v0.3 warnings	13–20% → ≤ 0.31%
05	Preregistered simulation criteria passed (v0.2 study · v0.3 study)	53/62 · 40/40

Sources: results/V02_SUMMARY.md and results/V03_SUMMARY.md in the repository. Row 03 uses the empirically calibrated configuration, not the anytime-valid one. Row 04 is a closed-loop simulation with illustrative degradation rates.

01 · EVIDENCE STATUS · AS OF OCTOBER 2026

WHERE THE WORK STANDS

STATUS	ITEM	DETAIL
Complete	**Simulation studies**	Two preregistered studies. 53 of 62 criteria (v0.2) and 40 of 40 (v0.3) passed; the failures are published alongside the passes.
Complete	**Reproducibility**	`tools/verify_results.py` re-derives every summary from raw data and checks SHA-256 hashes; `tools/check_provenance.py` verifies clean source trees and preregistered seed blocks.
Ready, not run	**Hardware pilot**	14 jobs, at most 320,000 shots, rehearsed on fake backends. Requires an IBM Quantum account and the account holder's authorization.
Invited, not started	**Independent validation**	Proposed 30-day, read-only observation study on existing calibration records. No institutional partnership or endorsement is implied.

02 · BACKGROUND

QUBITS DRIFT BETWEEN CALIBRATIONS

Superconducting qubits are calibrated on a schedule, but their parameters move in between. Klimov et al. (2018) reported that energy-relaxation times can fluctuate unpredictably by up to an order of magnitude in time and frequency, and traced the largest fluctuations to individual two-level-system defects. Burnett et al. (2019) found the same mechanism in transmons over multi-day runs and noted that it explains the need for frequent re-calibration.

Watching for drift is only useful if the warnings can be trusted. A monitor that re-tests after every batch against a fixed p-value or posterior bar will eventually alarm in almost any long session. E-values and e-processes, which stay valid at arbitrary stopping times, address this (Ramdas et al. 2023; Shin, Ramdas and Rinaldo 2022). qnp applies them to qubit monitoring. CaliScalpel and the DAQEC probe-deploy study approach the same drift from the calibration and error-correction side.

03 · CAPABILITIES

THREE THINGS THE PROFILER DOES

A

DETECTS INSTABILITY

Every estimate carries a provenance: *observed* (vendor calibration), *inferred* (from ordinary jobs) or *measured* (dedicated sequences), and series are never mixed. Two prepare-and-measure probe circuits added to a job become a measured readout-error series. Drift, threshold crossing and trend are each tested by a sequential e-process, and an alert needs statistical evidence, an effect of at least 10%, and persistence.

B

CONTROLS FALSE ALARMS

The monitor declares one session error budget ($\alpha = 0.10$) and splits it across channels, qubits, metrics and provenances. Where the null assumptions hold, Ville's inequality bounds the chance of any false alert at any monitoring time, for any session length. Each alert states its e-value, threshold and the kind of guarantee behind it: exact, Bayesian-marginal, approximate, or empirically calibrated.

C

KNOWS WHAT IT CAN'T TRUST

A T1 or T2 value is reported only if delay coverage, deviance, posterior-predictive and residual checks pass; otherwise the status is `insufficient_evidence` or `model_misspecification`. Forecasts abstain with machine-readable reasons such as `INSUFFICIENT_HISTORY`, `UNSTABLE_TREND` and `UNSUPPORTED_HORIZON`. Since v0.3, interleaved ground references flag `READOUT_DRIFT` during a sweep instead of letting it bend the estimate.

04 · RESULTS

FROM V0.1 TO V0.3, IN NUMBERS

The project's own audit of v0.1 found that 62% of stable sessions near threshold raised an alert, and that the T1 estimator was confidently wrong in 72% of fluctuating-T1 cases. The v0.2 study replaced per-check posterior bars with anytime-valid e-processes; v0.3 then fixed two of v0.2's preregistered failures. Each study used disjoint seeds for development, calibration and testing, and ran every method on identical data.

QUESTION (SIMULATION)	V0.1	V0.2	V0.3
Stable 8 h sessions with any false alert (qubit at 4.5%, threshold 5%)	0.70	0.010 av 0.068 cal	—
Stable 24 h sessions with any false alert	0.77	0.005 av 0.070 cal	—
×1.2 step detected within 6 batches (matched false-alert rate)	0.20	0.73	—
T1 confidently wrong when T1 fluctuates	0.72	0.00	0.00
T1 confidently wrong under slow readout drift in the sweep	—	0.255	**0.005**
False alert after adding 2 µs idle, hardware unchanged	—	1.00	**0.02**
False alert after adding two-qubit gates, hardware unchanged	—	0.43	**0.05**
80% forecast intervals containing the true crossing time	—	0.800	—

Table 2. "av" = anytime-valid thresholds; "cal" = the empirically calibrated configuration, which trades the theorem for an empirical guarantee. v0.3 rows use 100 runs (false alerts) or 200 fits (T1) per scenario. A dash means the version was not tested on that question. Sources: README; paper/paper.md §7 and §11.

05 · AGAINST STANDARD METHODS

COMPARED WITH SHEWHART AND CUSUM CHARTS

The studies compared the profiler with Shewhart and CUSUM control charts and with v0.1's rules re-thresholded, all on identical simulated data and with thresholds set to a matched false-alert rate.

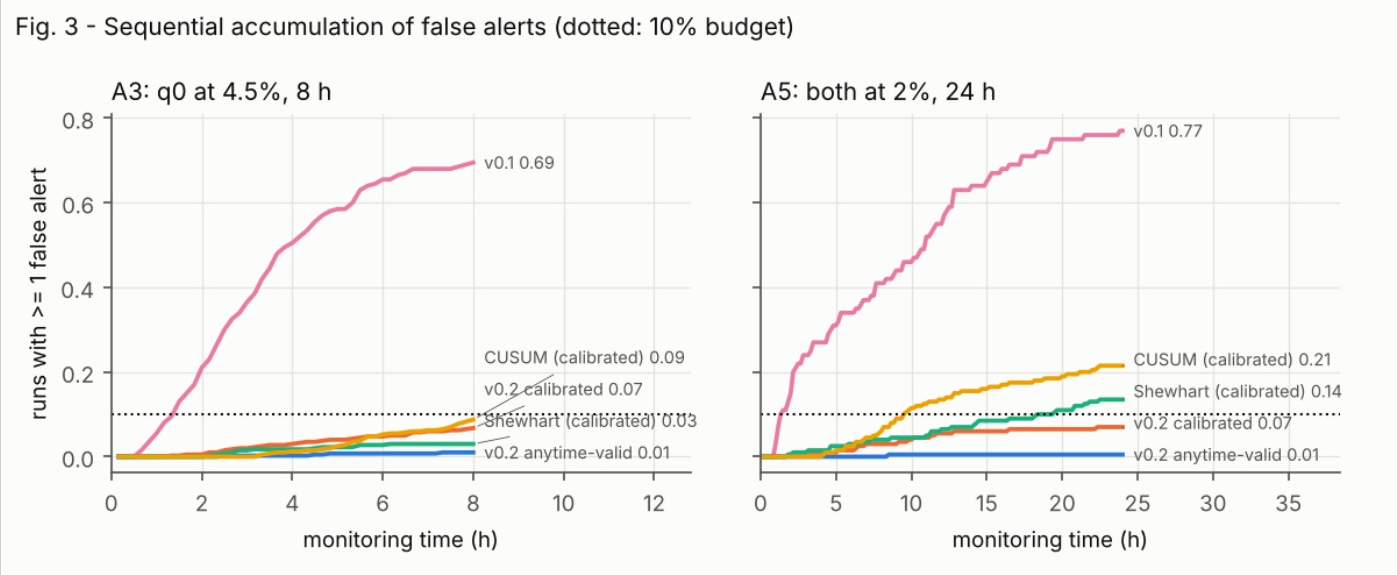


Fig. 3 (repository). False alerts accumulate over a session under v0.1's per-check bar and stay bounded under v0.2's e-processes. Charts calibrated on 8 h sessions exceeded their target over 24 h (Shewhart 0.135, CUSUM 0.215). Source: paper/figures/fig3_false_alert_accumulation.png.

WHERE IT IS BETTER

Longest median warning on gradual drift at a matched false-alert rate: 253 min, against 223 (Shewhart), 213 (CUSUM) and 203 (v0.1 re-thresholded). Fastest on moderate (×1.2) steps. False-alert control that does not depend on session length, which average-run-length charts do not offer.

WHERE IT IS NOT

A calibrated CUSUM detects the smallest (10%) steps more often within 24 batches (0.64 vs 0.52). No method detects a 10% step within 5 batches at this shot budget (0.09, against an original claim of 0.80). Stationary 10% fluctuations of a stable device still alarm in about half of runs.

06 · MEASUREMENTS IT REFUSES TO REPORT

CATCHING READOUT DRIFT INSIDE A T1 SWEEP

v0.2's model checks examine residuals along the delay axis. When a T1 sweep acquires its delays in order, a slow readout drift becomes a smooth function of delay and the fit silently absorbs it; in the v0.2 study that produced a confidently wrong T1 in 25.5% of fits. v0.3 acquires delays in random order and interleaves five ground references (prepare |0⟩, measure) whose outcome cannot depend on T1, so a homogeneity test on them measures drift directly.

SCENARIO (200 FITS EACH)	SEQUENTIAL	RANDOM ORDER	+ 5 REFS (V0.3)
AR(1) readout offset (v0.2 D11 conditions)	0.255	0.090	**0.005**
Slower AR(1), same sd	0.200	0.130	**0.025**
Linear offset ±1.5% across the sweep	0.370	0.110	**0.000**
Valid fits rejected (the cost)	1.5–2%	—	2.5–4%

Table 3. Fraction of fits reporting a confidently wrong T1 (status OK, truth outside the 95% interval). The five references add about a third to the shots of a 15-delay sweep. Source: results/V03_SUMMARY.md, Experiment H.

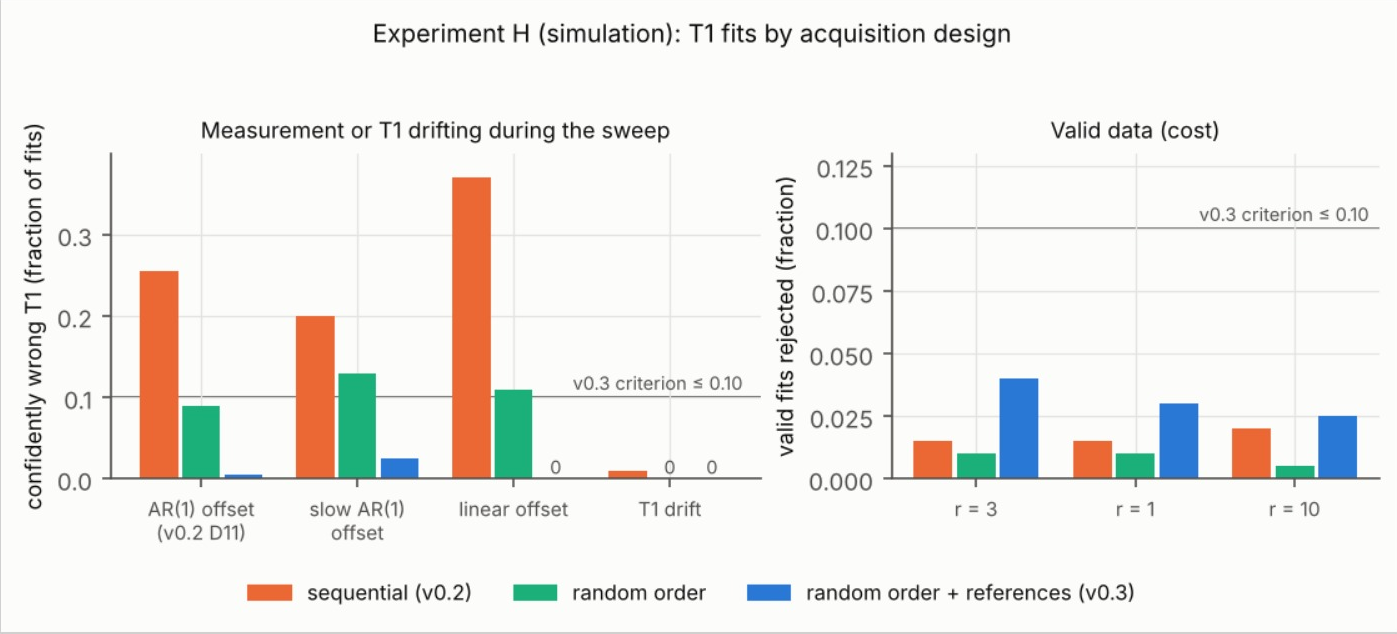


Fig. 9 (repository). The three sweep designs under measurement drift. Misspecifications caught earlier (fluctuating T1, 70/30 bi-exponential decay, short delay ranges) stayed caught in 400 of 400 fits. Source: paper/figures/fig9_sweep_designs.png.

07 · OPERATIONAL TEST

DO EARLIER WARNINGS HELP?

Experiment I simulates the operator's decision. Two qubits degrade through random gradual onsets and abrupt jumps at illustrative rates, on top of a stationary ±5% fluctuation of the readout error. A calibration costs one 10-minute batch and resets the qubit; a job fails if it runs while its qubit's true readout error exceeds 5%.

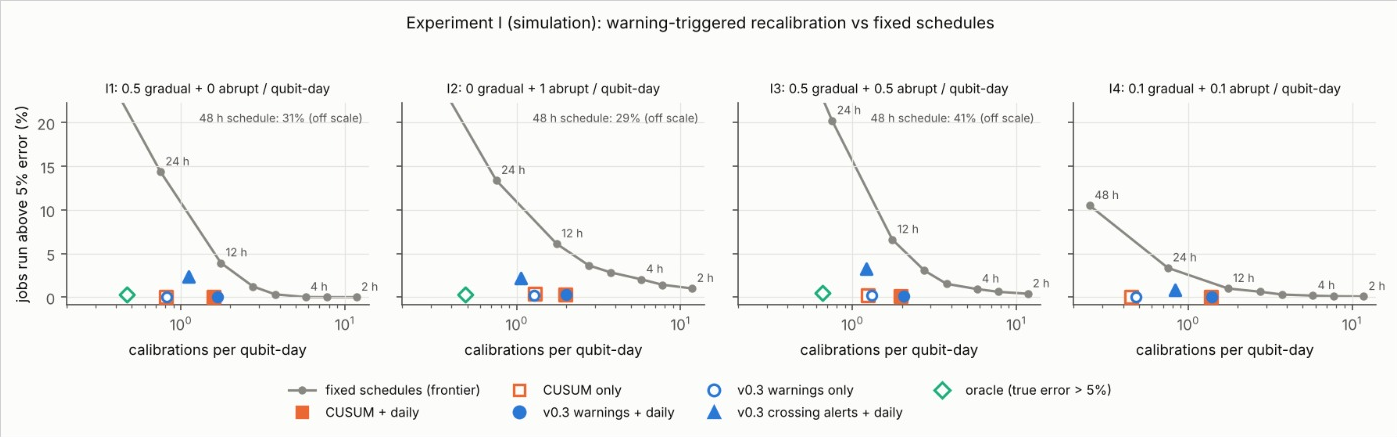


Fig. 10 (repository). Failed jobs against calibrations per qubit-day. Fixed schedules from every 48 h to every 2 h trace the frontier; monitoring policies sit below it. Source: paper/figures/fig10_operational_frontier.png.

13–20%

Jobs failed under a daily schedule alone (scenarios I1–I3)

≤ 0.31%

With v0.3 warnings added; 4.7–5.4 points under the frontier at equal cost

+0.5

Calibrations per qubit-day vs the daily schedule when degradation is daily; fewer only when it is rare

WHAT DID NOT HOLD UP

A calibrated CUSUM on the same probes did as well: failed-job differences were within 0.16 points, and the engine used 0.07–0.09 more calibrations per qubit-day in three of ten comparisons. The repository's own reading is that the gain comes from monitoring readout probes and acting early, not from the e-process. False triggers were 0.27–0.44 calibrations per qubit-day, and probe shots were not counted as a cost.

08 · WHAT IS NOT KNOWN

OPEN QUESTIONS

REAL HARDWARE

No real-device data exist. Batch independence, binomial counts and stable readout within a job are assumptions. A two-hour, 14-job pilot (at most 320,000 shots) is preregistered and rehearsed on fake backends.

OTHER NOISE

Leakage, coherent errors, crosstalk, 1/f dephasing and correlated multi-qubit errors are not simulated. A mild 90/10 bi-exponential decay still passes the model checks in 99.5% of fits.

OPERATIONAL VALUE

The closed-loop result uses chosen degradation rates and assumes recalibration on demand, which cloud users cannot do. Their intervention, avoiding or remapping a degraded qubit, has not been tested.

SCALE AND GUARANTEES

Families of two qubits were tested; the union-bound budget grows conservative with many qubits and metrics. The drift channel's guarantee is Bayesian-marginal, and the calibrated configuration's is empirical only. The full study ran in one computing environment.

09 · ENGAGEMENT

INDEPENDENT VALIDATION

FOR LABORATORIES AND AGENCIES

The project invites replication and a proposed 30-day, read-only observation study on existing calibration records or authorized exports. That phase would submit no jobs, change no device settings and claim no ground-truth detection; it can examine data quality, repeatability and agreement with recorded calibration changes. Researchers at national laboratories, NIST, universities and other facilities can review the collaboration checklist. No institutional partnership or endorsement is implied.

Proposal and checklist: docs/HARDWARE_VALIDATION.md. Contact: open an issue on the repository.

FOR RESEARCH GROUPS

MIT licence. Install with `pip install -e ".[qiskit]"`, or `".[dev]"` for tests, Cirq, IBM fake backends and figures. Cite via the repository's `CITATION.cff`, and also cite Qiskit, Qiskit Aer, Qiskit Experiments and Cirq.

Repository: github.com/topherchris420/quantum-noise-profiler

10 · QUICK START

RUNNING THE PROFILER

```
pip install -e ".[qiskit]"
python examples/demo_qiskit_sim.py

from qnp import DriftConfig, ErrorBudget, Profiler
from qnp.alerts import format_text
from qnp.measured import readout_probe_circuits

probes = readout_probe_circuits(physical_qubits,
            num_qubits=backend.num_qubits)
prof = Profiler(config=DriftConfig(
    budget=ErrorBudget(alpha_session=0.10,
                       n_qubits=5, n_metrics=2)))
prof.ingest_qiskit([*isa_workload, *probes],
                   sampler_job.result(), timestamp=ts)
alerts = prof.check()
print(format_text(alerts))
reports = prof.report()
```

Pass the executed (ISA) circuits so qubit indices are physical. Readout probes and workload series count as two metrics. Declare the monitoring family up front: a family that grows later voids the budget and sets `prof.family["exceeded"]`.

Alerts carry their e-value, threshold, α and guarantee type. `DriftConfig(log_e_offset=2.0)` is the study's calibrated configuration; its alerts are labelled `+calibrated`. `LegacyDriftConfig()` reproduces the frozen v0.1 rules.

Every reported number is recomputable: `python tools/verify_results.py` re-derives the summaries from raw data and checks SHA-256 hashes.

REFERENCES

Repository. topherchris420/quantum-noise-profiler: README.md, RELEASE_NOTES.md (v0.1 to v0.3).

paper/paper.md. v0.2 research report with the v0.3 follow-up (§11).

results/V02_SUMMARY.md. v0.2 criteria and failures.

results/V03_SUMMARY.md. Experiments G, H, I; pilot P status.

docs/HARDWARE_VALIDATION.md. Observation study and collaboration checklist.

paper/figures/. Figs. 3, 9 and 10 reproduced as published.

Klimov et al. 2018. Fluctuations of energy-relaxation times in superconducting qubits. PRL 121, 090502.

Burnett et al. 2019. Decoherence benchmarking of superconducting qubits. npj Quantum Information 5, 54.

Ramdas, Grünwald, Vovk, Shafer 2023. Game-theoretic statistics and safe anytime-valid inference. Statistical Science 38(4).

Shin, Ramdas, Rinaldo 2022. E-detectors: a nonparametric framework for sequential change detection. arXiv:2203.03532.

Fang et al. 2024. CaliScalpel: in-situ qubit calibration integrated with surface-code QEC. arXiv:2412.02036.

DAQEC-Benchmark. Drift-aware QEC dataset with IBM hardware validation. Zenodo record 18045662.

Repository figures are reproduced from the files listed; outside sources supply context only and were not used to evaluate the tool. Nothing in this report is evidence about real quantum hardware.